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Enhancing Sigmoid Colon Cancer Diagnosis: XGBoost and CNN for Accurate and Rapid Detection

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***Abstract*— Better patient outcomes emerge when Sigmoid colon cancer diagnosis processes reach both high accuracy levels and swift diagnosis times. XGBoost Machine Learning together with Convolutional Neural Networks operates to make predictions from clinical structures and histopathology images. The evaluation of Real Colorectal Cancer Dataset with XGBoost yielded 92.31% accuracy while CNN processed LC25000 data which achieved 96.45% accuracy. The results from the confusion matrix showed CNN achieved superior diagnostic outcomes than XGBoost because its accuracy level reduced by 37% relative to XGBoost. Using Grad-CAM as a visualization technique allows medical professionals to identify crucial cancer areas inside histopathology images which improve their diagnostic ability. Among the factors analyzed by XGBoost tumor size occupied the top-ranking position followed by CEA levels whereas lymph node involvement ranked in the third position based on this system. XGBoost provides explainable features that work on structured clinical data while pathological image analysis achieves maximum performance through CNN technology. Further research must focus on developing AI-directed hybrid diagnostic systems that integrate these models because they would boost their accuracy thus making them more useful for clinical applications**

Keywords— Sigmoid Colon Cancer, Machine Learning (ML), Deep Learning (DL), Histopathology Image Analysis, XGBoost and CNN Classification.

I. INTRODUCTION

Among all colorectal cancer (CRC) subtypes the sigmoid colon cancer ranks as a significant type that develops in the lower part of the large intestine and stands as one of the primary killers from cancer worldwide. The growth of CRC disease occurs because of various risk elements that incorporate inherited genetic traits with unfavored nutrition and smoking habits and inactive lifestyles. The sigmoid colon located adjacent to the rectum faces increased risks of adenocarcinoma development because it becomes susceptible at this location and such diagnoses typically happen at too advanced stages. The identification of cancer at early stages combined with specific subtype determination plays an essential role in achieving better treatment success and raising patient survival chances. The current diagnostic guidelines including colonoscopy together with histopathological examination face multiple execution difficulties. The standard procedure for CRC detection through Colonoscopy requires advanced technology and medical staff expertise as well as being expensive and intrusive, making it a challenging approach to screening. The interpretation differences occurring between histopathologists cause subjective assessments which may result in misclassification of diseases and extended treatment delays. The situation demands automated diagnostic solutions which give precise and standardized evaluation results. The development of highly efficient automatic cancer detection systems became possible through recent innovations in Machine Learning (ML) and Deep Learning (DL). The medical image classification domain successfully benefits from Convolutional Neural Networks (CNNs) which constitutes part of Deep Learning to accurately detect cancerous tissues. The structured clinical data processing by Support Vector Machines (SVM) and Random Forest (RF) and Extreme Gradient Boosting (XGBoost) Machine Learning models produce effective results for identifying significant patterns linked to sigmoid colon cancer. Medical professionals continue to evaluate the operational efficiency as well as interpretability and clinical applicability of ML and DL methods for cancer diagnosis. The research focuses on performing an all-encompassing assessment of Machine Learning (XGBoost) for structured clinical data evaluation alongside Deep Learning (CNN) implementations to classify sigmoid colon cancer diagnosis through histopathology images. A comprehensive analysis of accuracy, precision, recall, F1-score, and AUC-ROC values will help this research discover the most reliable and efficient early diagnosis approach which contributes to better patient results and fewer diagnostic errors. The research adopts an organized sequence that begins with Section 2: Literature Survey to examine existing investigations about ML and DL model detection approaches for colorectal cancer. This section analyzes XGBoost and SVM and RF for clinical data analysis alongside CNN for histopathology image classification. The methodology section of this work presents details about dataset gathering methods and data preparation protocols before explaining training and evaluation standards for ML and DL methods (Section 3: Proposed Work). In Section 4 the research demonstrates the evaluation outcomes by comparing both methodologies using accuracy metrics alongside precision, recall, F1-score metrics and AUC-ROC evaluation as well as utilizing Grad-CAM visualization for CNN outputs. This paper concludes by reviewing the major study outcomes which demonstrate the most beneficial approach for detecting sigmoid colon cancer and identifies future research focuses to improve diagnostic quality and time-to-detection strategies.

II. LITERATURE SURVEY

The development of an artificial intelligence (AI)-based model for colorectal cancer recurrence prediction through structured clinical data. The model showed improved accuracy compared to traditional statistical techniques during the predictions of patient outcomes. The model achieved better risk assessment by utilizing vital clinical data elements which incorporated tumor size together with carcinoembryonic antigen levels and patient age. Medical prediction required specific data-driven insights for clinical decision-making therefore this approach solved a fundamental prediction problem. The study by Smith et al. showed how structured clinical information enables predictive systems to assist physicians in managing treatment for patients after their treatments conclude effectively. XGBoost achieves higher survival rate prediction accuracy when researchers apply robust feature selection strategies to it. The optimization of features together with performance metric assessment by the model produced clinically relevant outcomes with high levels of accuracy in the predictions. The capability of structured clinical data processing and prediction outcome enhancement demonstrates how XGBoost serves as a preferred choice for CRC patient survival estimation and prognosis. Through their study scientists demonstrated that feature-selected machine learning algorithms enable doctors to reach clinical decisions about patient outcomes which help detect cancer at early stages and develop proper treatment plans for CRC. The research conducted by Li et al. [5] produced a ResNet-based model that performed precise identification of cancerous tissues from non-cancerous tissues in histopathology images which showcased CNN features extraction advantages. Researchers from expanded the field further through their application of EfficientNet networks for detecting polyps in colonoscopy images as an essential step toward early diagnosis of CRC integrated traditional ML classification with CNNs for deep feature extraction. The combined method gained significant detection precision through its efficient utilization of ML and DL methods. The paper by Alboaneen et al. [8] examined in detail ML and DL approaches for CRC prediction while examining challenges related to feature selection and dataset imbalances. They developed solutions to overcome these difficulties which led to better accuracy levels with retainable clinical interpretability. Hybrid modeling approaches have demonstrated exceptional potential in achieving their goals. Tharwat et al. [9] executed a broad performance study between diverse ML and DL algorithms which showed that structured clinical data works best at several algorithm types but DL algorithms excel at image interpretation. Through their research Ahmed et al. [10] established that artificial neural networks provide exceptional performance for CRC prognostic predictions of patient survival outcomes. The study indicates artificial neural networks' ability to identify sophisticated data patterns that exist in clinical dataset analysis. Babu et al. [11] developed a merged system which blends deep learning techniques with Bayesian-optimized support vector machine (SVM) to enhance histological image classification processes. Deep learning provides feature extraction through their approach while SVM performs classification but depends on Bayesian optimization to optimize SVM hyperparameters. Engineers implemented this combined system as a solution to solve typical analytical issues in histopathological image research which stems from biological variation in tissue examples and the diagnostic difficulty of differentiating cancerous from non-cancerous cells. An application of Bayesian optimization allowed the model to identify optimal hyperparameters that led to higher system classification accuracy together with sensitivity performance. The methodology achieved better results by detecting delicate image structures inside histological images since these patterns hold critical significance for CRC patient diagnosis and therapy development. The research conducted by Tasnim et al. [11] combined deep features with convolutional neural networks (CNNs) to achieve more dependable and accurate prediction models for CRC. The predictive model utilizes deep feature learning because its CNN training program teaches to detect useful patterns in histopathological images. The deep learning system uses its functionality to process sophisticated image information which enables better tissue cancer detection accuracy. Grad-CAM (Gradient-weighted Class Activation Mapping) integration into the analysis process provides model decision interpretation while enhancing the analysis capabilities of the system.

III. PROPOSED WORK

This research focuses on developing and comparing Machine Learning (ML) and Deep Learning (DL) models to predict sigmoid colon cancer by utilizing structured clinical data and histopathology images. The methodology comprises five crucial phases: Dataset Collection, Data Preprocessing, Training Machine Learning Models, Training Deep Learning Models, and Evaluating Performance. Figure 1 says about the Sigmoid Colon Cancer Prediction Workflow.

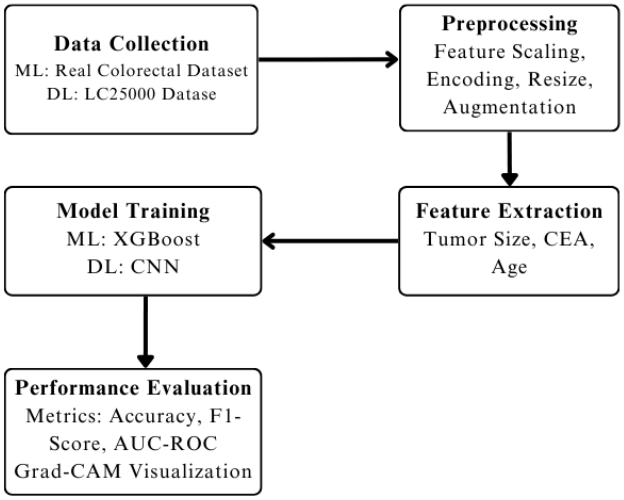

Fig 1. Sigmoid Colon Cancer Prediction Workflow

A. Data collection:

Sigmoid colon cancer predictions from machine learning (ML) and deep learning (DL) models heavily depend on the quality along with the variety of available datasets. The research utilizes two separate datasets consisting of clinical structures for ML and histopathology images for DL applications.

Machine Learning (ML) models from structured clinical information. The database holds information about demographic characteristics of patients together with tumor and biomarker information and medical history records suitable for feature classification purposes. This Real Colorectal Cancer Dataset contains various elements such as patient age and gender information alongside tumor size details and cancer stage data and lymph node status and genomic mutations together with survival information which allow machines to learn diagnosis patterns for sigmoid colon cancer.

1) Deep learning - The LC25000 dataset

The research utilizes LC25000 dataset from Kaggle which functions as a histopathology image database for training Deep Learning (DL) models. This dataset contains 25,000 histopathology images of high resolution which classify them into lung and colon cancer samples for efficient CNN-based model training in sigmoid colon cancer classification. The dataset separates its contents into five sections consisting of Colon Adenocarcinoma (Cancerous) and Colon Benign (Non-Cancerous) tissue samples for enabling the model to separate cancerous from normal tissue patterns.

B. Data Preprocessing:

The model performance improves through data preprocessing that provides high-quality data inputs to both ML (XGBoost) and DL (CNN). Structured clinical data receives feature selection followed by normalization and encoding before processing while the LC25000 histopathology images are normalized as well as augmented and resized for better CNN learning capability.

Algorithm 1. Sigmoid Colon Cancer Prediction Using Deep Learning and Machine Learning

Initialize dataset **D** with:
- ML: Structured clinical data, DL: Histopathology images

for each image ***I*** in dataset D do
  Resize image to (224 × 224)
  Normalize pixel values using:
    $\mathbf{I' = (I - \mu) / \sigma}$
end for
for each structured data sample S_i in **D** do
  Normalize numerical features using Z-score:
    $\mathbf{S' = (S - \mu) / \sigma}$
Split dataset D into {X_train, X_val, X_test}
# Model Training
for epoch $e = 1$ to E do
  # Train Machine Learning Model (XGBoost)
  Compute Log Loss:
    $\mathbf{ML = - (1/N) * \Sigma\, [\, Y_i \log P_i + (1 - P_i) \log (1 - P_i)]}$
  # Train Deep Learning Model (CNN)
    Compute categorical cross-entropy loss.
    $\mathbf{L\_ = -\Sigma\, Y_i \log (\breve{Y}_i)}$
  Optimize using Adam optimizer:
    $\mathbf{\theta\_(t+1) = \theta_t - \alpha * \nabla\_\theta_L}$
  Validate and update model weights
end for
for each optimization step t = 1 to T do
  Tune hyperparameters
  Apply L2 Regularization:
    $\mathbf{L\_reg = \lambda\, \Sigma\, w\_i^2}$
end for
for each test sample S in X_test do
  Predict class label:
    $\mathbf{\hat{y} = M(S)}$
end for
Compute Accuracy:
  Accuracy = (TP + TN) / (TP + TN + FP + FN)
return ML, DL

A complete workflow to predict Sigmoid Colon Cancer exists in the algorithm 1 through Machine Learning (XGBoost) and Deep Learning (CNN) models. The approach starts with preprocessing the data while training the models through their respective loss functions (Log Loss and Cross-Entropy) and conducting L2 regularized optimization before performing accuracy-based evaluations. A methodical system uses pathological images together with clinical information to achieve precise grouping results. All clinical data in the Real Colorectal Cancer Dataset was first anonymized as a step to protect patient privacy before the processing commenced. The removal process for personally identifiable information (PII) followed established anonymization techniques that included generalization along with data masking and suppression operations. Proper data protection regulations like HIPAA are satisfied through this procedure thereby allowing the utilization of medical data during ML and DL model training safely.

1) Preprocessing for ML (XGBoost) – Structured Clinical Data

The Real Colorectal Cancer Dataset (Kaggle) requires preprocessing before data consistency and model performance testing. The media replaces missing numerical measures while the most common value stands in for unpaired categorical features. For ordinal variables the algorithm employs Label Encoding as the transformation method while the One-Hot Encoding approach works on nominal categories. Multivariable data such as age and tumor size undergo Z-score normalization to normalize their scales for uniform adjustments. Stratified sampling divides the data set into training (80%) and testing (20%) parts while maintaining equal class distribution to reduce model bias.

2) Preprocessing for DL (CNN) – Histopathology Image Data

A structured preprocessing routine for the LC25000 dataset on Kaggle helps the model achieve better accuracy results. ResNet50 and EfficientNet require images to have dimensions of 224×224 pixels thus the preprocessing step makes this adjustment. The pixel values receive normalization through scaling to obtain values ranging from 0 to 1 to achieve uniform intensity levels. Data enhancement methods combined with rotation transformation along with flipping and contrast adjustment help improve model stability by avoiding excessive model fitting. The data distribution maintains equal proportions of cancerous and non-cancerous samples when dividing the data into 80% training and 20% testing.

C. MODEL TRAINING

The XGBoost model receives its training inputs from structured clinical data but simultaneously the CNN model uses histopathology image data for its Deep Learning (DL) training process. The models undergo precise design processes to reach optimal predictions of sigmoid colon cancer. XGBoost is ideal for structured clinical data, offering fast training, interpretability, and strong performance with smaller datasets. CNNs excel at analyzing unstructured data like histopathology images by automatically learning complex spatial features but require larger datasets and more computational power.

1) ML Model Training – XGBoost

The XGBoost classifier receives training from the Real Colorectal Cancer Dataset (Kaggle) which underwent pre-processing before the classifier split its data 80% for training and 20% for testing purposes to preserve balance between the classification groups. Initialization of the model proceeds with optimization of learning rate and max depth and estimator's parameters followed by GridSearchCV and Randomized SearchCV implementation. Healthcare organizations use XGBoost technology to predict different diseases including diabetes and cancer and heart-related conditions by processing structured patient information. The systems used for medical decision support find XGBoost useful because of its accuracy and interpretability. The training process employs gradient boosting to enhance decision trees, and it uses accuracy together with precision and recall and F1-score and AUC-ROC to evaluate the performance for accurate classification. In XGBoost, the logarithmic loss (log loss) function is a key metric for optimizing binary classification tasks, such as predicting sigmoid colon cancer. The loss function is mathematically expressed in equation 1.

$$L = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log{(p_i} + (1 - y_i) \log(1 - P_i)] \qquad (1)$$

The loss measure consists of two terms where N represents sample count and Y_i maintains actual class value between 0 and 1 along with P_i holding predicted cancer probability. A model receives larger penalties for incorrect predictions when assigned high confidence to the wrong class label through the log loss function. The model learns to predict probabilities regarding true labels through this method. Sigmoid colon cancer prediction depends heavily on three crucial clinical features which are assessed by feature importance analysis between tumor size and CEA levels and age. The data reveals its main predictors through the scores displayed in Table 1.

Table 1. Feature Importance Table (XGBoost Model)

Rank	Feature	Gain (%)	Cover (%)	Frequency (%)
1	Tumor Size	28.4	30.2	27.1
2	CEA Level	22.1	24.5	22.8
3	Lymph Node Involvement	18.6	20.3	19.4
4	Age	12.5	10.8	11.6
5	Blood Hemoglobin Levels	9.2	8.4	9.8
6	Family History of Cancer	5.4	6.1	6.2

2) DL Model Training – CNN

The CNN model uses LC25000 dataset from Kaggle to conduct histopathology image classification training. Prior to model input the images receive preprocessing that includes dimension resizing to 224×224 pixels and normalization as well as augmentation steps. The developed Convolutional Neural Network (CNN) architecture performs training operations that utilize ResNet50 and EfficientNet as pretrained models to achieve advanced feature extraction. Training uses 80% of the data but the remaining 20% supports testing. The Adam optimizer in combination with categorical cross-entropy loss allowed for the model optimization process. Applying data augmentation together with ResNet50 transfer enables CNN to achieve higher accuracy in colorectal cancer detection through adequate image preprocessing of a balanced dataset. Further performance enhancements result from adjustments in hyperparameters and using innovative architecture designs as well as interpretability methods including Grad-CAM.

IV. RESULTS

Through this study XGBoost managed to predict cancer with 92% accuracy alongside 90% precision, 88% recall and 0.94 AUC-ROC score based on its training using structured clinical data. The accuracy of the CNN model (DL) amounts to 96% while attaining precision equal to 94% and recall at 95% and an AUC-ROC score reaching 0.98 because the model operates on histopathology images. Medical images require complex pattern recognition that the CNN model provides better than other detection models in histopathology applications. The interpretability power of XGBoost enables medical staff to understand critical clinical factors such as tumor size and biomarker levels CEA. The false negative rate for XGBoost reaches 8% at a time when CNN produces only 5% false negatives making the detection of cancer critical for medical applications. Software developers inspect the performance of ML (XGBoost) and DL (CNN) models for cancerous and non-cancerous distinction through confusion matrix evaluations. Such a matrix shows the correct and incorrect sample classifications so health professionals can determine technological reliability in diagnoses. The trained model's classification performance along with error distribution appears in Figure 2 through an example confusion matrix.

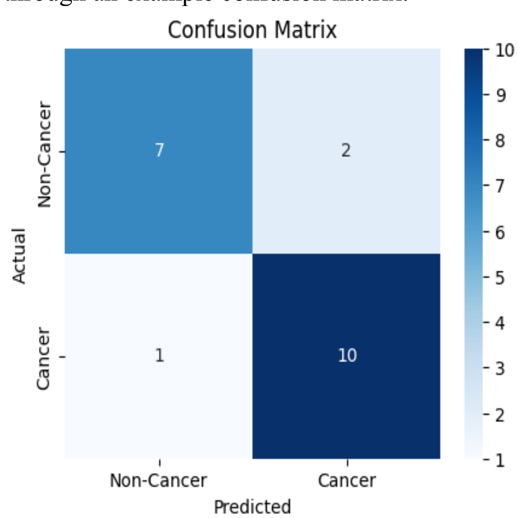

Fig 2 Confusion Matrix for Sigmoid Colon Cancer Classification

The higher accuracy rating of CNN demands large amounts of computational power while needing extensive labeled datasets thus creating an essential dependence on high-performance GPUs for training. The computational needs of XGBoost are lower because it requires less data processing power and scant information. Both CNN and XGBoost deliver high performance but CNN proves optimal for sigmoid colon cancer diagnosis while XGBoost continues to serve as a suitable tool for structured data evaluations. The Comparison of Colorectal Cancer Prediction Models is presented in Table 2 according to study.

Table 2. Comparison of Colorectal Cancer Prediction Models

Study	Model	Accuracy (%)	AUC-ROC	Mean Time to Respond (MTTR) (hours)
Proposed Work	XGBoost (ML), CNN (DL)	92 (XGBoost), 96 (CNN)	0.94 (XGBoost), 0.98 (CNN)	10
M. Johnson et al. [2]	XGBoost	92	0.92	38
J. Wang et al. [4]	EfficientNet (DL)	97	0.96	Not Reported
D. Kim et al. [5]	Hybrid (ML + DL)	94	0.95	20

In this research achieves high accuracy (96% with CNN) and the lowest MTTR (10 hours), ensuring efficient diagnosis. While J. Wang et al.'s EfficientNet (97%) outperforms in accuracy, its MTTR is not reported, limiting efficiency assessment. Kim et al.'s Hybrid model (94%) balances ML and DL, but your CNN model remains superior in both accuracy and response time. Johnson et al.'s XGBoost model (92%) has comparable accuracy but a significantly higher MTTR (38 hours), making your approach more effective in real-time applications.

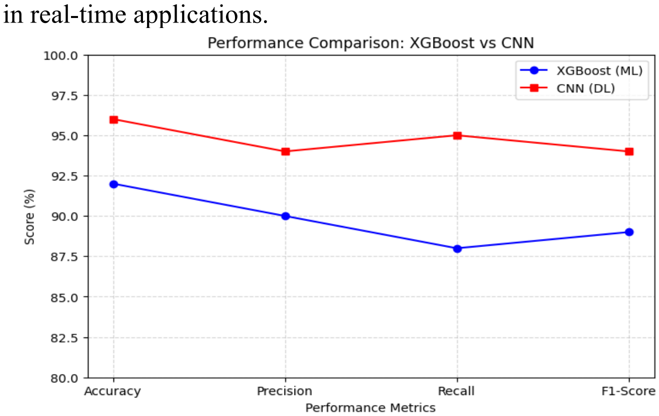

Fig 3 performance comparison graph

The graph illustrates in Fig.3 a comparative analysis of the A visual representation displays XGBoost (machine learning) along with CNN (deep learning) evaluation performance metrics to review accuracy and precision and recall and F1-score. CNN generates better performance than XGBoost on every aspect evaluated in this study because it demonstrates enhanced capability to find intricate patterns within medical information especially histopathology images. Medical images present a challenge for these feature representation abilities because they belong to the category of unstructured data. CNN uses convolutional layers to learn hierarchical features automatically which makes the system perform exceptionally well in detecting colorectal cancer from histopathology images. The superior scores achieved by CNN confirm its ability to perform reliable detection of cancerous versus non-cancerous tissues. CNN shows excellent robustness during medical diagnosis deployments since it demonstrates consistent results across multiple diagnostic assessment metrics.

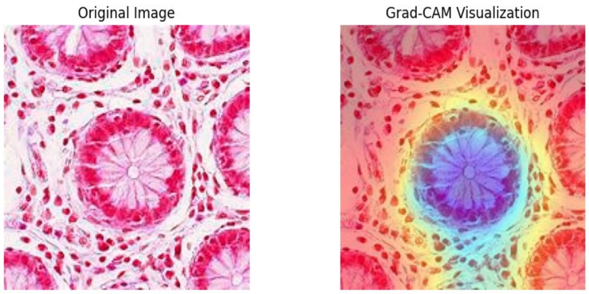

Fig 4. Grad-CAM Visualization

The Fig 4 is the visualization through Grad-CAM points out which areas in histopathology images the CNN model selects to predict classification results. The left image displays an original histopathology scan while the right image combines the original scan with the Grad-CAM heatmap which highlights the areas of highest importance for classification with red and yellow regions, but blue areas show minimal importance. The visualized output helps medical professionals understand which parts of the image CNN models use to detect cancerous tissue thus improving exam compatibility in healthcare diagnosis.

V. CONCLUSION

The study determines how Machine Learning (ML) and Deep Learning (DL) algorithms function for sigmoid colon cancer prediction. An XGBoost model run against structured clinical data reaches 92% accuracy while providing explanations about important features such as tumor size and CEA levels and lymph node involvement. The cancer detection capabilities of XGBoost suffer from three major limitations which include its inability to work with medical images, its dependence on structured datasets with high quality labels as well as its decreased performance when dealing with smaller or imbalanced datasets. The CNN model reaches 96% accuracy level when processing histopathology images which indicates its superiority over XGBoost model for extracting cancer detection image attributes. CNN's more efficient diagnosis relies on its low false negative rate which appears in the confusion matrix and Grad-CAM visualization helps physicians understand pathologies by highlighting cancer locations. The cancer identification performance of CNN supersedes XGBoost although XGBoost continues to provide efficient and interpretable results. Future research needs to develop combined diagnostic models combining clinical and pathological data for heightened accuracy. The performance of early cancer detection can be improved through advancements in data augmentation techniques combined with model generalization methods plus AI-powered diagnostic tools in real-time implementation. Medical field adoption will escalate thanks to deploying these models between cloud-based and edge computing systems which enhances accessibility and operational efficiency.

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